

September

SPECIAL EVENTS

PHASES OF THE MOON

	Full Moon	New Moon
2008	September 15	September 29
2009	September 4	September 18
2010	September 23	September 8
2011	September 12	September 27
2012	September 30	September 16

THE PLANETS

2008: September 6–7

Mercury and Mars appear about five Moon-widths apart in the western evening sky.

2008: September 11

Mercury is at greatest evening elongation, magnitude 0.2.

2008: September 11

Venus and Mars appear less than a Moon-width apart in the western evening sky.

2010: September 19

Mercury is at greatest morning elongation, magnitude –0.1.

2010: September 21

Jupiter is at opposition, magnitude –2.9. At midnight, it is visible to the south from northern latitudes and to the north from southern latitudes.

2011: September 3

Mercury is at greatest morning elongation, magnitude 0.0.

Opposition and elongation are explained on [page 14](#)

The stars of northern fall and southern spring are now in place as Capricornus and Aquarius move to center stage. On the 23rd, or occasionally the 22nd, of the month, the Sun moves from the northern to the southern celestial sky. When it crosses the celestial equator, day and night are of equal length. In the weeks ahead, the northern hemisphere nights lengthen and the southern ones shorten.



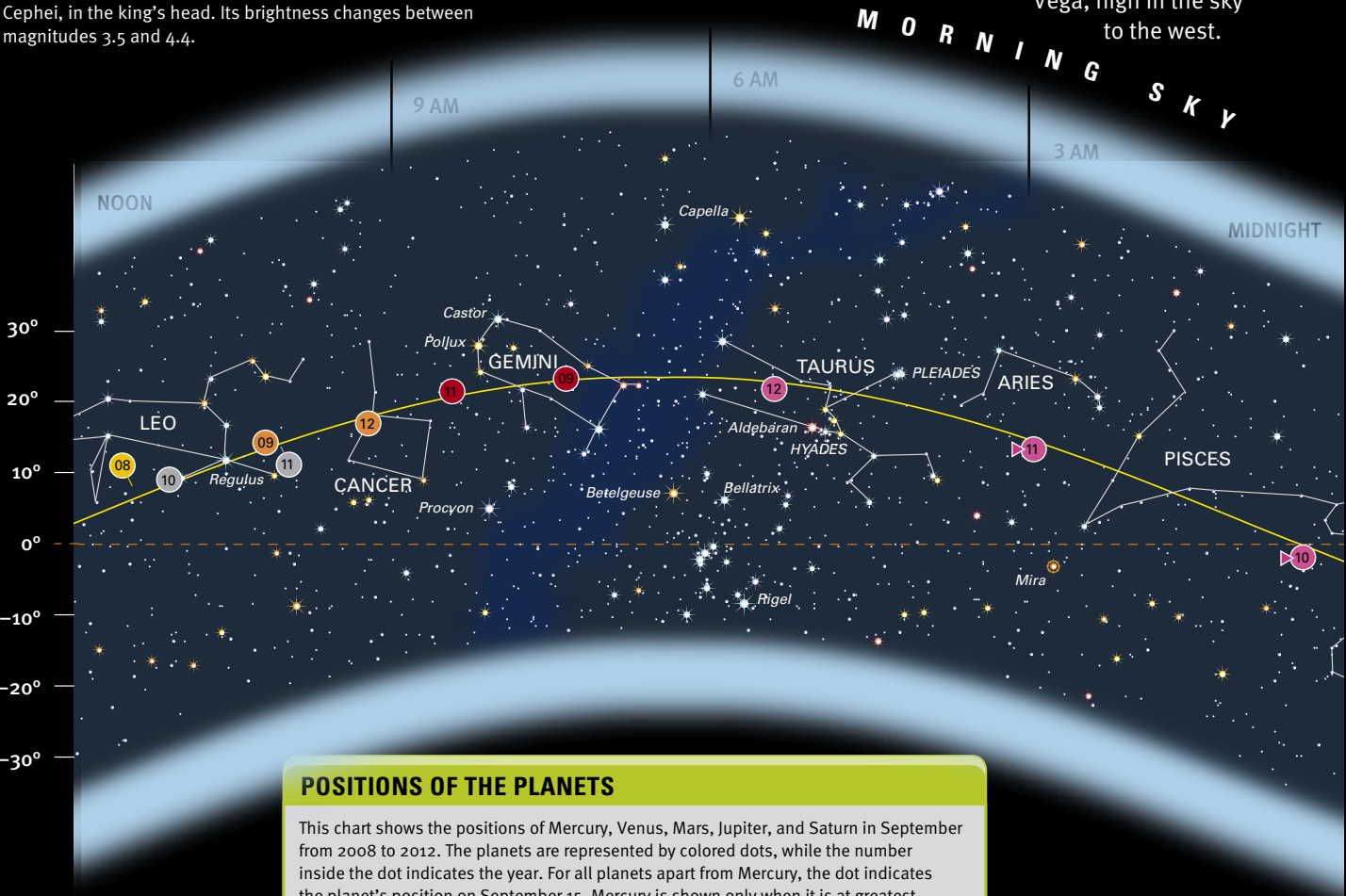
Cepheus

The head of Cepheus, the mythical king of Ethiopia, is at lower left. The constellation is not prominent but is worth seeking out for Delta Cephei, in the king's head. Its brightness changes between magnitudes 3.5 and 4.4.

NORTHERN LATITUDES

Capricornus and Aquarius are at either side of due south. For those at high northern latitudes, September is the best time to see Capricornus, the smallest zodiac constellation. Cygnus remains high overhead and is joined by the flying horse, Pegasus. The Summer Triangle stars, Deneb (Cygnus), Vega (Lyra), and Altair (Aquila), remain in view to the west, but Pegasus's presence in the east announces the arrival of fall.

To the north, Ursa Major is below Polaris, while Cepheus is above. Its star Delta (δ) Cephei is the prototype of the Cepheid variables. As the star pulsates, its brightness shifts over a five-day cycle. Only two bright stars are seen to the north: Capella (Auriga), the sixth-brightest of all, is close to the northeast horizon. Just brighter is Vega, high in the sky to the west.



POSITIONS OF THE PLANETS

This chart shows the positions of Mercury, Venus, Mars, Jupiter, and Saturn in September from 2008 to 2012. The planets are represented by colored dots, while the number inside the dot indicates the year. For all planets apart from Mercury, the dot indicates the planet's position on September 15. Mercury is shown only when it is at greatest elongation (see p.14)—for the specific date of elongation, refer to the table to the left.

Mercury Venus Mars Jupiter Saturn

EXAMPLES

09 Mars's position on September 15, 2009

09 Saturn's position on September 15, 2009. The arrow indicates that the planet is in retrograde motion (see p.14)